

€LvÈ[fmwu5UF[¢g {it |û~ßV1.0 •oNö<¾<j<ôf Nf

{,N zà i,•đ

1.1 ~yvî€ìfo

→w@u5UF^LN v„_ë• SÑ\Uÿ [¢g Vâ→ v„{it N Wù<-b N:O N h8_ÃzPN%oR>v„,í%o•~Äb •èR 0 O ~ßv„[¢g {it e¹_ [XW(eHs±ONN 0 Wù<-Thg ••0
u5UF[¢g O\N:O N N m^•9€ Iÿ• v„,í%o•heh•ÿ Qvg R¡•(‘İvôc¥_qTÍu(b7OSšÆTÆTÁrL_bÆa0 q6€ ÿ O ~ß[¢g {it —bN4<øY c b ÿ N°TXmAR““ÿ
g,|û~ß• •Çep[WS 0 fz€ýS v„e¹_ ÿ ^.R©u5UFO N [žs°[¢g Vâ→ v„|¾~ÆS {it ÿ cÐSGetOSg R¡|4^sTÆ•Ð„%eHs±0

1.2 |û~ßvîh

g,|û~ßN;%o•[žs°NâN vîh ÿ

1. [žs°[¢g Vâ→ v„Qhu T}Thg {it
- NÍQe€LR0y»€Lv„[ÆethchH{it
- Wù<-•Ů^!•ýŽ*N €ýR>•ÄO0
- €LN SÑ\U•İ_„%oÄR
2. cÐO>fz€ýS v„Wù<-N € h8OS|û
- ~Óg„S <þz OS|û
- R -6kµWù<-•ýŽ*
- KPI’İS € h8
3. g„^úwâ<Æ^“N <Ýg/^“e/d’OS|û
- NŞTÁwâ<Æ^“
- h QÆ<Ýg/^“
- ^LN wâ<Æ^“
4. cÐO>epcnR g•N Q³{Ve/c
- [žeðepcnNê~hvØ
- •<R¿R g•b¥TJ
- •İ~<Êe~„<f
5. e/c Y %oÒ,rSOO\N gC—P{it
- Y ~šgC—PcšR6
- •è•è•èSOO\
- díO\|i<j•ýŽ*

1.3 • u(f Vô

{, 1 ~u

g,|û~ß• u(NŽu5UFO N v„[¢g Vâ– {it ý e/c NâN %oÖ,rý

- O N {it TXÿ |û~ß'M•nN Qh\@_{it
- O N Oá`o{it
- YW™ <¢– {it
- epcn~ß<ijR g•
- [¢g N;{iy Vâ– {it N € h8<ÄOO
- Vâ– b TX{it
- Wù<-NûRijR 'M
- KPI€ h8bg^L
- [¢g N°TXÿ eâ^8[fn`N]âO\k°_U
- <pz [fn`
- <Ýg/gâ–
- •ĩ~<Ëe-

1.4 e‡hcf Vô

- g,<¾<ij<ôf NfmµvÖ|û~ßv„NâN e¹—by
- |û~ßetOSg¶g„<¾<ij
 - RÝ€ýj!WW<æ~Æ<¾<ij
 - epcn^“~Ög„<¾<ij
 - c¥Sâ%oÄf <¾<ij
 - [%oQhg:R6<¾<ij
 - `€ýO S <¾<ij
 - mK<ÖN •è•re¹hH

{,NÆZà |û~ßg¶g„

2.1 b€g/g¶g„

g,|û~ß'Çu(s°NâS Qhh b€g/g¶g„ÿ QER R)u(N'<ij{—TÆ_@g Rit _ð0

- RMzib€g/h ý
- Next.js 16ÿ App Routerÿ ý g Rijin2gÓhFg¶
 - React 19ÿ u(b7uL—b^“
 - TypeScript 5ÿ |{W<[%oQh_ SÑ
 - Tailwind CSS 4ÿ SÝ[PS CSShFg¶
 - shadcn/ui~ÄNö^“ÿ šØ•('IUI~ÄNö

- T zib€g/h ý
- Next.js API Routesÿ g RijizíAPI
 - Supabaseÿ PostgreSQLÿ ý N'epcn^“g Rij
 - Drizzle ORMÿ |{W<[%oQhORM

{, 2 ~u

b€g/• WˆO Rzÿ

1. Qhh JavaScript/TypeScriptÿ –MON[fn`b g,
2. g Rjzin2gÖcÐSG™–\O`€ÿTÆSEO
3. |{W<[%QhQİ\ •Ð`Leö• <i
4. N'g Rj–MON•Ð~ôb g,

2.2 |û~ßg¶lg,,Vp

|û~ß'Çu(–İQxv,,N \Bg¶lg,,<¾<¡ÿ

ˆhs°\Bÿ Presentation Layerÿ ÿ

- Webu(b7uL—b
- yûR`zi• 'MuL—b
- {,N e¹–Æb c¥Sã

N Rj\Bÿ Business Layerÿ ÿ

- u(b7<¤<Ág Rj
- N Rj•;•'Y t
- epcnh!ŠÆ•lcb
- m`^oc•• g Rj

epcn\Bÿ Data Layerÿ ÿ

- Qs|ûW<epcn^“
- [ûÆa[XP`g Rj
- • [Xg Rj
- eâ_x[XP`

2.3 j!WWg¶lg,,

|û~ßc Rÿ€ÿR R N:NâN h8_Ãj!WWÿ

h8_Ãj!WWÿ

1. <¤<Ác^gCj!WWÿ Authÿ
2. u(b7{it j!WWÿ Userÿ
3. [¢g {it j!WWÿ Agentÿ
4. Wù<-N- _Ãj!WWÿ Trainingÿ
5. € h8<ÄÖ0j!WWÿ KPIÿ
6. wâ<Æ^“j!WWÿ Knowledgeÿ
7. AIR©bKj!WWÿ AIÿ
8. epcnR g•j!WWÿ Analyticsÿ

e/d'j!WWÿ

1. e‡Nö[XP`j!WW
2. m`^o• wâj!WW

3. eâ_x[<ij!WW
4. [šeoNûRij!WW

2.4 •è•rg¶g,,

|û~ße/c N'zĩ•è•rÿ 'Çu([¹VhS •è•re¹_ÿ

- u Nšs~Xfÿ
- ^"u(g R;Vh~Æ•▪
 - epcn^"N;NîY R6
 - [ùŒa[XP" g R;
 - CDNR • g R;

- _ Sñs~Xfÿ
- g,W0_ Sñg R;Vh
 - _ Sñepcn^"
 - j!bßg R;

- è•rmAz ÿ
1. Nâx g,^úbSS
 2. •Pĭg,^úc"•
 3. g RjnÚR" fôe°
 4. Pe^•hÂgâšŒÁ

{,N zà RŸ€ýj!WW<¾4<ij

3.1 Nê^hvØj!WW

Nê^hvØf/|û~ßv,,h8_ÃQeSäÿ N:u(b7cÐO>epcni,%oÈTŒ_ëcwfíO\0

RŸ€ý<ôf ÿ

1. Qs•.c h i,%oÈ
- Vâ~ N°ep~ß<ij
- Wù<[Œb s‡
- € h8^sWGR
- •ĩ~Y t 'İ

2. _...RžN<~ycÐ"
- _...Y t]âSU
- _...[ih8Q...[¹
- Ss\ R0g NûR;
- |û~ß• wâ

3. epcn•<R¿Vp^h
- g R;•(İ•<R¿

{, 4~u

- Wù<-•Û^|fò~¿
- € h8b ~éR ^
-]åSUY t ~ß<j

4. _ëcwfÍO\QeSã

- _ë• R ^ú]åSU
- _ë• SÑ•w€ h8
- _ë• gâw b¥TJ

uL—b<¾<jÿ

- TÍ^” _ ^ \@• ‘MN T \O^U
- SarG_ Oá`o\Uy:
- SibÖbýc’^•
- epcn[žëöR7e°

3.2 [¢g {it j!WW

[¢g {it j!WW[žs°[¢g N°TXv„Oá`o{it TÆr¶` •ýŽ*0

RŸ€ŸR ^hÿ

1. [¢g Oá`o{it

- Wúg,Oá`o_UQe
- Oá`o• •Oie9
- Y4PíN O {it
- by’Í[üQe[üQú

2. Wù<—6kµ•ýŽ*

- Qe€LWù<—6kµ
- \—RMWù<—6kµ
- W(\—cÐSG—6kµ
- rızÉN \—R\$[š

3. r¶` {it

- W(€Lr¶`
- <Öu(r¶`
- y»€Lr¶`
- r¶` SØfô<°_U

4. €LOM{it

- U.N-[¢g
- U.T [¢g
- ~Ä••
- N;{i

epcn[Wkµ<¾<jÿ

{, 5 ~u

- Wúg,Oá`oÿ YÓT 0]âS÷0 Qe€Leåg
- \—OMOá`oÿ €LOM0 b@V^Vâ— 0 IGb¥[ùŒEa
- Wù<-Oá`oÿ Wù<—6κμ0 [Œb <pz 0 € h8b ~é
- r¶` Oá`oÿ W(€Lr¶` 0 R ^úeö•ô0 fôe°eö•ô

3.3 Wù<-N-_Ãj!WW

Wù<-N-_Ãf/|û~ßv„h8_ÃRŸ€ýj!WWÿ e/d'[¢g N°TXv„b ••SÑ\U0

<pz {it RŸ€ýÿ

1. <pz R ^úN ••‘
 - <pz Wúg,Oá`o
 - <pz Q...[¹• c’
 - \—bVprGN O
 - SÑ^ r¶` {it

2. <peö[%c’

- <peöQ...[¹• •‘
- [fN`eö••<¼•n
- _ÄOî• Oîh <°
- <peöc’^•

3. hHO<Qs€T

- [žb hHO<~Ñ[š
- hHO<%oãg•<ôf
- vøQswâ<Æ”pc¥

4. [fN`•De™

- e‡hc•De™N O
- %oÆ~’•De™{it
- Y •ê”pc¥_ u(

[fN`•ýŽ*RŸ€ýÿ

1. [fN`•Û^i<°_U
 - <pz [Œb •Û^i
 - <peö[fN`r¶`
 - [fN`eö••~ß<ij

2. [Œb s‡~ß<ij

- N*N°[Œb s‡
- Vâ— [Œb s‡
- R |[Œb s‡

3. b ~é<Ä[š

- <pz mKšŒb ~é

- [ždí<ÄR
- ~üt <Ä[š

4. SÍ™^e6—Æ
- <pz <ÄN÷
 - [fN`_Ä_—
 - e9•Ü^ú<®

3.4 KPI€ h8j!WW

KPI€ h8[žs°[¢g ~éeHv,,İS <ÄO0ÿ e/c Y ~ô^|€ h8OS|û0

e¹hH{it Rÿ€ÿÿ

1. € h8e¹hHR ^ú
- e¹hHWúg,Oá`o
 - € h8Thg <¼•n
 - • u(N°TX• bé
 - e¹hHr¶` {it

2. c h gC´í<¼•n
- c h `y[šNI
 - gC´ÍR 'M
 - <ÄR S:•ô
 - R R `y<¼•n

3. <ÄR h QÆ[šNI
- <ÄR {l~šR R
 - {l~šcl•ô<ôf
 - y:O<SÂ€

4. e¹hHrHg,{it
- rHg,S†Sò<°_U
 - rHg,[ùkÔ
 - rHg,VpnÚ

€ h8bg^LRÿ€ÿÿ

1. [šg € h8SÑ•w
- •êR`Thg € h8
 - bKR`SÑ•w€ h8
 - € h8NûRjR 'M

2. <ÄR epcn_UQe
- N;{<ÄR
 - •ê<ÄR ep
 - T N<N'<Ä

{, 7~u

- |û~ß<ÄR

3. ~Ógœ<ij{—IG`;
- R gCR ep<ij{—
- c`T ~ß<ij
- •<R¿[ùkÔ

4. b¥TJu b [üQú
- N*N°€ h8b¥TJ
- Vâ— IG`;b¥TJ
- •<R¿R g•b¥TJ

3.5 <Ýg/^"j!WW

<Ýg/^"cÐO>h QÆS v„[¢g ^" {Tj!g•ÿ ^.R©[¢g _ë• TÍ^" [¢b7•ĩ~0

R |{{it OS}üÿ

1. U.RM<Ýg/
- N§TÁT`ˆ<â<Ýg/
- OÃ• m;R`ˆ<Ýg/
- N SU_ [ü<Ýg/
- e/NØ•ĩ~<Ýg/

2. U.T <Ýg/
- rimAgã<â<Ýg/
- • cb•ˆ<Ýg/
- •(ˆĩ~<Ýg/
- O•u(c [ü<Ýg/

3. b•<ÉY t
- b•<Éc¥e6<Ýg/
- [%bšlÿ• <Ýg/
- %ãQ³e¹hH<Ýg/
- VP<¿•ß•Û<Ýg/

4. rykŠW:fo
- Y`OÃg •ô<Ýg/
- „,PGeã<Ýg/
- VIP[¢b7<Ýg/
- }ˆ%` ÅQµ<Ýg/

Rÿ€ÿryp¹ÿ

1. W:foS R |{
- c N RlW:foR |{
- c [¢b7|{W<R |{

{, 8 ~u

- c •ĩ~|{W<R {

2. fz€ýd }"

- Qs•.<íd }"
- <íNld }"
- h {~{[•

3. O•u(˘k!~ß<¡

- pí•è<Ýg/c˘L
- O•u(•<R¿R g•
- eHgæ<ÄO0

4. eHgæSí™^e6—Æ

- g eH/eàeHh <°
- e9•Û^ú<®
- rHg,•ínă

3.6 wă<Æ^“j!WW

wă<Æ^“e/d'[¢g v„eă^8[fN`TCE•ĩ~%oă{Tý f/O N wă<Æv„l%oMĂ^sSđ0

Q...[˘{it OS|ûÿ

1. NŢĂWă<Æ

- NŢĂWúg,Oá`o
- NŢĂ%oĂh<SÂep
- O•u(e˘lŒ<ôf
- ^8%oĂ•ĩ~%oă{T

2. ^LN wă<Æ

- ^LN e?{VIŒ%oĂ
- ^LN h QÆ%oĂf
- ^LN •<R¿R˘˘
- zᐅTÁ[ùkÔR g•

3. dÍO\mAz

- |û~ßdíO\c SW
- N RjmAz <ôf
- lèa N<˘ycĐ"
- ^8%oĂ• <i%oĂ••

4. ^8%oĂ•ĩ~

- FAQ•î{T^“
- •ĩ~R |{h {~
- %oăQ³e˘hH"pc¥
- vøQswă<ÆQs€T

{, 9 ~u

wâ</Æ|{W<e/c ÿ

1. e‡g,wâ</Æ

- ~e‡g,Q...[¹

- [l̃e‡g,••

- Markdowne/c

2. VprG<ôf

- b*Vp<ôf

- mAz Vpy:

- y:a Vp\Uy:

3. %/Æ~eYz

- díO\o y:%/Æ~

- Wù<-%/Æ~

- hHO<<²%ã%/Æ~

4. e‡hc–DNö

- PDFe‡hc

- Worde‡hc

- Excel~hh<

3.7 AIR©bKj!WW

AIR©bKcDO>fz€ýS e/c ÿ ^.R©[¢g fôšØeHW0%ãQ³•ĩ~0

RŸ€ýW:foÿ

1. •ĩ~<Êe-R g•

- •ĩ~|{W<</ÆR+

- SŸVàR g•^ú<®

- %ãQ³e¹hHc~fP

- vøQshHO<_ u(

2. <Ÿg/O S ^ú<®

- ^h•¾e¹_O S

- <íl cª•ž^ú<®

- N*~S Œ et

- eHgæ<ÄO0

3. Wù<-Q...[¹c~fP

- [fN`Q...[¹c~fP

- ..._1s~_n_:S

- •Ů–6<pz ^ú<®

4. epcnm [ßR g•

{, 10~u

- epcn•R¿%ã<û
- _ ^8~„f
- Ö S ^ú<®

b€g/[žs°y

1. WúNŽY'<íŠ j!W<
 - 'Çu(QH•ÜLLMb€g/
 - e/c Y j!W<R cb
 - c ~íO S cĐy:<í

2. N N e‡t %ã
 - Y •η[ù<Ýt %ã
 - N N e‡Qs€T
 - a Vp<ÆR+

3. mA_ Tí^"
 - [žëðTí^"•"Qú
 - bS[Wg:eHgæ
 - cĐSGu(b7OSšÆ

4. Y •η[ù<Ý
 - [ù<ÝStSòOÝc
 - N N e‡•Ɓ•/
 - fz€ý•ý•î

3.8 •ĩ~<Êe-j!WW

•ĩ~<Êe-^.R©[¢g _ë• [šOM•ĩ~^vb~R0%ãQ³e¹hH0

<Êe-mAz <¾<¡y

1. •ĩ~cĩ•đ•"Qe
 - •ĩ~R |{• bé
 - <æ~Æcĩ•đ•"Qe
 - vøQsOá`o`eQE
 - VprG–DNöN O

2. Alfz€ýR g•
 - •ĩ~|{W<<ÆR+
 - Qs•.Oá`ocĐSÖ
 - SÝVàc`t R g•
 - N%`íz ^¡<ÄO0

3. %ãQ³e¹hHc`fP
 - h Q/Æ%ãQ³e¹hH
 - S†SòvøO<hHO<

- <Ýg/^ú<®
- læa N<~y

4. vøQswâ<ÆQs€T
- vøQsN\$TÁwâ<Æ
 - vøQsdÍO\mAz
 - vøQse?{V%oÄ[š

<Êe-{|W<R |{ÿ

1. N\$TÁ•ĩ
- N\$TÁ•('ĩ
 - N\$TÁO•u(
 - N\$TÁ%oÄh<

2. rimA•ĩ
- SÑ•'^ö•ß
 - rimAgââ<â
 - {~e6_ ^8

3. U.T •ĩ
- •cb•'
 - ~ôOîg Rj
 - b•<ÉY t

4. •&b7•ĩ
- v{ _U•ĩ
 - e/NØ•ĩ
 - O TX•ĩ

3.9]åSU{jt j!WW

]åSU{jt [žs°•ĩ~v„•ßŽ*Y t ÿ xnOÝkĪN*•ĩ_—R0Y¥U„%oãQ³0

]åSUmAz <¾<jÿ

1.]åSUR ^ú
- •ĩ~Wúg,Oá`o
 - [¢b7Oá`oQs€T
 - }`%z ^!h <°
 - ••#N°R 'M

2. R m>Y t
- •êR`R m>%oÄR
 - bKR`R m>Æ et
 - •lm>Y t
 - SORŽ•€<÷

3. •Û^!•ßŽ*
- Y t •Û^!fôe°
- [¢b7• wă
- •...eö~„<f
- P~RžcĐ”

4. [Œ~Ó_Rhc
- %ăQ³e¹hH<°_U
- [¢b7náa ^!_
- _RhcR |{
- wă<ÆI%mA

]ăSU|{W<ÿ

1. [¢b7b•<É
- g R¡b•<É
- •(“!b•<É
- rimAb•<É

2. •ĩ”T”<â
- N§TÁT”<â
- e?{VT”<â
- díO\T”<â

3. SOO\Y t
- •è•è•èSOO\
- •Dn•SOŒ
- •ĩ”SG~§

4. Q...•èSOO\
- wă<ÆQqN«
- ~İšŒENªmA
- Wù<-SOR©

3.10 •"Ri{it j!WW

•"Ri{it [žs°e6Qee/Qúv„<°_UR g•ÿ e/d‘~İ„%Q³{V0

RŸ€ýf Vöÿ

1. e6k><°_U
- <¢SUE6k>
- O TXQEP<
- QvNÖe6Qe

2. NØk><°_U

- b g,e/Qú
- •9u(b¥•
- 'Ç•-NØk>

3. b g,h8{—
- N°R>b g,
 - •Ð„%b g,
 - b€g/b g,

4. R)m!R g•
- kÚR)<i{—
 - QÀR)R g•
 - R)m!s‡~ß<i

b¥^hRŸ€ÿÿ

1. eåb¥^h
- _Seåe6e/IG`;
 - ^8Nœf cÐ”
 - 'İp¹QslèN<^y

2. Thb¥^h
- Th^!e6e/IG`;
 - s^kÔR g•
 - •<R¿Vp^h

3. g ^!IG`;
- g ^!e6e/f ~Æ
 - ~„{—bg^L`ÅQμ
 - _ ^8R g•

4. •<R¿R g•
- e6e/•<R¿Vp
 - ~Ôg„R g•
 - ~„mKR g•

{,VÛzà epcn^“<¾<ij

4.1 epcn^“i,•đ

|û~ß'Çu(PostgreSQLQs|ûW<epcn^“ÿ • •ÇSupabaseN'g R;çÐO>bX{i0 epcn^“<¾<ij•u_ª{,N f _ SŸR ŷ xnOŸepcnN •ô`'TÆ[Æet`'0

epcn^“ryp¹ÿ

1. N'zibX{iÿ šØSiu(OŸ—œ
2. •êR`Y Nÿÿ epcn[‰Qh
3. ^L~š[‰Qhÿ ~Æ|'!gC—P

4. [žëö<¢– ÿ epcnT ke

4.2 h8_Äepcn^h

u(b7^h(users)<¾<jÿ
 - id: u(b7U/N h <Æ(UUID)
 - email: v{ _U•@{±(U/N)
 - password_hash: [ÆEx TÈ^
 - display_name: f>y:T yð
 - role: u(b7%œÖ,r
 - user_type: u(b7|{W<
 - company_id: b@\\^O N (Y •.)
 - status: •&b7r¶|^
 - last_login_at: g T v{ _Ueö•ô
 - created_at: R ^úeö•ô
 - updated_at: fôe°eö•ô

O N ^h(companies)<¾<jÿ
 - id: O N U/N h <Æ(UUID)
 - name: O N T yð
 - industry: b@\\^LN
 - team_size: Vâ– %œÄj!
 - contact_name: €T|ûN°YÓT
 - contact_phone: €T|ûu5<Ý
 - plan: <¢– YW™ (basic/pro/flagship)
 - plan_start: YW™ _YËeö•ô
 - plan_end: YW™ ~Ög_eö•ô
 - trial_end_at: <Öu(~Ög_eö•ô
 - status: O N r¶|^
 - created_at: R ^úeö•ô
 - updated_at: fôe°eö•ô

[¢g ^h(agents)<¾<jÿ
 - id: [¢g U/N h <Æ(UUID)
 - company_id: b@\\^O N (Y •.)
 - name: [¢g YÓT
 - employee_id:]âS÷
 - hire_date: Qe€Leåg
 - position: €LOM(U.N-[¢g /U.T [¢g /~Ä••/N;j)
 - training_stage: Wù<–6kµ(Wúx@/U.N-/U.T /•Û–6/rizËN \—)
 - status: W(€Lr¶|^ (W(€L/y»€L/<Öu)
 - created_at: R ^úeö•ô
 - updated_at: fôe°eö•ô

<pz ^h(courses)<¾<jÿ

- id: <þz U/N h <Æ(UUID)
- company_id: b@\\^O N (Y •.)
- title: <þz h ~
- content: <þz Q...[¹
- category: <þz R |{
- cover_image: \ —bVprGURL
- duration: <þz eö••
- status: SÑ^ r¶|^
- created_at: R ^úeö•ô
- updated_at: fôe°eö•ô

KPIe¹hH^h(kpi_schemes)<¾<iÿ

- id: e¹hHU/N h <Æ(UUID)
- company_id: b@\\^O N (Y •.)
- name: e¹hHT yð
- description: e¹hHcĭ•ð
- metrics: € h8c h (JSON)
- weights: gC'Í'M•n(JSON)
- period: € h8Thg
- status: e¹hHr¶|^
- created_at: R ^úeö•ô
- updated_at: fôe°eö•ô

<Ýg/^“^h(phrase_library)<¾<iÿ

- id: <Ýg/U/N h <Æ(UUID)
- company_id: b@\\^O N (Y •.)
- category: <Ýg/R |{
- scenario: • u(W:fo
- content: <Ýg/Q...[¹
- tags: h {~(JSON)
- use_count: O•u(klep
- effectiveness: g eH`^<ÄR
- created_at: R ^úeö•ô
- updated_at: fôe°eö•ô

4.3 epcnQs|û<¾<j

N;%•[ŽOSQs|ûÿ

1. u(b7N O N ŷ Y [ùN Qs|û

- N N*O N g Y N*u(b7
- N N*u(b7\\^NŽN N*O N

2. [¢g N O N ŷ Y [ùN Qs|û

- N N*O N g Y N*[¢g
- N N*[¢g \\^NŽN N*O N

3. <þz N O N ÿ Y [ùN Qs|û
- N N*O N g Y N*<þz
- N N*<þz \^NŽN N*O N

4. KPIe¹hHN O N ÿ Y [ùN Qs|û
- N N*O N g Y N*€ h8e¹hH
- N N*e¹hH\^NŽN N*O N

5. <Ýg/N O N ÿ Y [ùN Qs|û
- N N*O N g Y N*<Ýg/
- N N*<Ýg/\^NŽN N*O N

Qs|û~!g_ÿ
- Y •.~!g_OÝ<Áepcn[ÆEt`
- ~\$€TR -d{Vue
- •oR -dg:R6

4.4 }" _ <¾<ij

}"_ u(NŽO S gâ<â`€ýÿ

1. N;•.}"_
- b@g ^h•ýg N;•.}"_
- UUID|{W<N;•.

2. Y •.}"_
- company_id[Wkµ}"_
- user_id[Wkµ}"_

3. N Rj}"_
- emailU/N }"_
- statusr!` }"_
- created_ateö•ô}"_

4. Y T }"_
- (company_id, status)Y T }"_
- (company_id, created_at)Y T }"_

4.5 epcn[%oQh<¾<ij

[%oQhcªe½ÿ

1. ^L~\$[%oQh{Vue(RLS)
- kĭN*^hT/u(RLS
- u(b7Sê€ý<¿•îg,O N epcn
- service_role~Ö•ÇRLS

2. epcnh!šŒ

- g Rjzëpcnh!šŒ
- |{W<[%Qh~!g_
- N Rj%ÄR h!šŒ

3. eOa epcnOÝbα

- [Æx TÊ^ [XP"
- eOa [Wkμ•1eO
- R [ÆO •"

4. dÍO\|i<j

- Qs•.dÍOleå_x
- epcnSøfô<°_U
- _^8dÍO\TJ<f

{,N"zà c¥Sã<¾<j

5.1 c¥Sã%Äf

|û~ß'Çu(RESTful API<¾<i~!h<ÿ •u_ªNåN %Äf ÿ

HTTPe!|Œ<iNIÿ

- GET: gâ<â•Dn•ÿ ^B{ldíO\
- POST: R ^ú•Dn•ÿ —^^B{l
- PUT: fôe°•Dn•ÿ ^B{ldíO\
- PATCH: •èR fôe°ÿ ^B{ldíO\
- DELETE: R –d•Dn•ÿ ^B{ldíO\

URL<¾<i%Äf ÿ

- •Dn•T <íY ep_b_
- \B~\$Qs|ûn fp
- gâ<âSÄep•Çnä
- rHg,S÷RM•

TÍ^"h<_ %Äf ÿ

- b RÝTÍ^"•ÖVpëpcn
- • <iTÍ^"•ÖVp• <iOá`o
- R ~uTÍ^"S T+R ~uOá`o
- ~ßN HTTPr¶\` x

5.2 <α<Ác¥Sã

v{ _Uc¥Sãÿ

- URL: POST /api/auth/login

```
- <÷IBSÂëpy
{
  "email": "u(b7•@{±",
  "password": "u(b7[Æx "
}
- b RŸTÍ^"y
{
  "success": true,
  "token": "JWTNärl",
  "user": {u(b7Oá`o}
}
- Y1•%TÍ^"y
{
  "error": "v{ _UY1•%",
  "detail": "•@{±b [Æx • <i"
}
```

lèQÆc¥Sãÿ

- URL: POST /api/auth/register

```
- <÷IBSÂëpy
{
  "email": "u(b7•@{±",
  "password": "u(b7[Æx ",
  "companyName": "O N T yð",
  "displayName": "f>y:T yð"
}
- b RŸTÍ^"y
{
  "success": true,
  "userId": "u(b7ID",
  "trialEndAt": "<Õu(~Óg_eö•ð"
}
```

f·SÖProfilec¥Sãÿ

- URL: GET /api/auth/profile

- <÷IBY4ÿ Authorization: Bearer {token}

```
- b RŸTÍ^"y
{
  "user": {u(b7Oá`o},
  "company": {O N Oá`o},
  "planEnd": "YW™ ~Óg_eö•ð"
}
```

5.3 N R;ic¥Sã

[¢g {it c¥Săy

- R `hgă<ây GET /api/agents
- <æ`Ăgă<ây GET /api/agents/:id
- R ^úy POST /api/agents
- fôe°y PUT /api/agents/:id
- R –dÿ DELETE /api/agents/:id

<pz {it c¥Săy

- R `hgă<ây GET /api/courses
- <æ`Ăgă<ây GET /api/courses/:id
- R ^úy POST /api/courses
- fôe°y PUT /api/courses/:id
- R –dÿ DELETE /api/courses/:id

KPlē!hHc¥Săy

- R `hgă<ây GET /api/kpi-schemes
- <æ`Ăgă<ây GET /api/kpi-schemes/:id
- R ^úy POST /api/kpi-schemes
- fôe°y PUT /api/kpi-schemes/:id
- R –dÿ DELETE /api/kpi-schemes/:id

<Ýg/^"c¥Săy

- R `hgă<ây GET /api/phrase-library
- R ^úy POST /api/phrase-library
- fôe°y PUT /api/phrase-library/:id
- R –dÿ DELETE /api/phrase-library/:id

5.4 R ~uc¥Să

R ~uSĂepÿ

- page: ~ux y Nî1_ YĖ
- limit: kî~uep'ÿ žØ<²20

R ~uTÍ^"y

```
{
  "data": [...],
  "total": `ep'Ī,
  "page": _SRM~u,
  "limit": kî~uep'Ī,
  "totalPages": `;~uep
}
```

5.5 • <iY t

~ßN • <iTÍ^"h<_ y

{, 20 ~u

```
{
  "error": "• <ij{W<",
  "detail": "<æ~ÆOá`o",
  "code": "• <iNãx "
}
```

^8%Á• <iX ÿ
- 400: <÷IBSÂep• <i
- 401: g* <α<Á
- 403: eàgC–P
- 404: •Dn•N [XW(
- 500: g RjVhQ...è• <i

{,Qmzà [%oQh<¾<j

6.1 <α<Á[%oQh

<α<Ág:R6ÿ
1. JWTNärL<α<Á
- NärLg eHg <¾•n
- NärLR7e°g:R6
- NärLdα• e/c
2. [Æx [%oQh
- [Æx _: ^!%o•IB
- [Æx TÊ^ [XP""(bcrypt)
- [Æx [šg fôcbcĐ"
3. v{ _U[%oQh
- v{ _U~s‡–PR6
- _ ^8v{ _UTJ<f
- Ÿ <¾Y v{ _U{it
4. O <Ý{it
- O <Ý•...eöc§R6
- O <Ý^vSÑ–PR6
- O <Ý[%oQhlè•

6.2 gC–Pc§R6

%oÒ,rgC–P<¾<jÿ
1. admin%oÒ,r
- |û~ß{it TX
- Qh\@'M•ngC–P
- b@g epcn<ç•î

2. enterprise_admin%Ò,r

- O N {it TX
- O N 'M•ngC–P
- g,O N epcn<¿•î

3. enterprise_manager%Ò,r

- [¢g N;{i
- Vâ– {it gC–P
- g,Vâ– epcn<¿•î

4. staff%Ò,r

- fn• [¢g
- Wúg,díO\gC–P
- N*N°epcn<¿•î

gC–Ph!ŠÆ\B~Šÿ

1. •ïu1~\$gC–Pc\$R6

- ~u—b<¿•îgC–P
- API<¿•îgC–P

2. RŸ€ý~\$gC–Pc\$R6

- RŸ€ýSi%Á`
- díO\Sibg^L`

3. epcn~\$gC–Pc\$R6

- epcnSi%Áf Vð
- epcndíO\gC–P

6.3 epcn[%Qh

epcnOÝb¤cªe½ÿ

1. O •“[%Qh

- HTTPS_:R6R [Æ
- <ÁNfg eHŠÆ<Á
- [%QhY4•è<¾•n

2. [XP“[%Qh

- eOa epcnR [Æ
- epcn•1eOY t
- [%QhR –dg:R6

3. Y Ný[%Qh

- [šg •êR`Y Ný
- Y NýepcnR [Æ

- _ W0Y Ný[XP''

4. [i<¡•ýŽ*
- dÍO\eaâ_x<°_U
 - epcnSØfô•ýŽ*
 - _ ^8^LN:TJ<f

6.4 [%oQh{Vue

[%oQh{Vue'M•nÿ

1. Q...[¹[%oQh{Vue(CSP)
- • g,bg^L–PR6
 - Y •è•Dn•–PR6
 - Q...€T• g,y•kb

2. •èWß{Vue(CORS)

- QA<_ WßT 'M•n
- <÷lBe¹lÖ–PR6
- Qí<Ád: ^&c\$R6

3. [%oQhY4•è

- X-Frame-Options
- X-Content-Type-Options
- Strict-Transport-Security

{,N zà ` '€ý<¾<¡

7.1 RMzi``'€ýO S

O S cªe½ÿ

1. Nâx R Rr

- •iu1~\$R+R Rr
- ~ÄNö~\$R+R Rr
- {,N e¹^"R Rr

2. aÖR •}

- VprGaÖR •}
- ~ÄNöaÖR •}
- •iu1aÖR •}

3. • [X{Vue

- —Y` •Dn•• [X
- APITí^"• [X
- Service Worker• [X

4. g Rjzïñ2gÓ
- ™—\Og Rjzïñ2gÓ
 - SEOSĚY}
 - `€ýcÐSG

7.2 T zĩ`'€ýO S

O S cªe½ÿ

1. epcn^"O S
- }" _ O S
 - gǎ<âO S
 - •Ɓc¥l`'M•n

2. • [X\B<¾<¡j
- Redis• [X
 - Q...[X• [X
 - • [X{Vue

3. _ keY t
- € eödÍO\ _ ke
 - m^`o— R
 - [šëŎŨRj

4. ^vSŃY t
- ^vSŃ<÷IBY t
 - •Ɓc¥Y u(
 - • •}WG^a

7.3 •Q~ÜO S

•Q~ÜO S cªe½ÿ

1. CDNR •
- —Y^ •Dn•CDN
 - VƁrGCDN
 - Qht R • „p¹

2. S<•)O •"
- GzipS<•)
 - BrotliS<•)
 - VƁrGS<•)

3. <÷IBO S
- <÷IBT ^v
 - <÷IBSÖm^
 - <÷IBS»'Í

7.4 Sîbî\U`'<¾<j

bî\U<¾<jÿ

1. j!WWS g!lg,,
 - RÿËÿj!WWrîzË
 - c¥Sãh QÆ~BN
 - ON€&T <¾<j
2. _@g Rj~„uY
 - g Rj•¹uLn fp
 - • OáSO<@h QÆ
 - g RjSÑs°~„uY
3. l4^sbî\Ue/c
 - eàr¶` <¾<j
 - O <Y Y •è[XP”
 - epcn^“<ûQ™R y»

{,Qkzà mK<Ö<¾<j

8.1 mK<Ö{Vue

mK<Ö\Bk!ÿ

1. SUQCmK<Ö
 - Qýep~SR+mK<Ö
 - ~ÄNó~SR+mK<Ö
 -]âQwQýepmK<Ö
2. –Æb mK<Ö
 - API–Æb mK<Ö
 - epcn^“–Æb mK<Ö
 - g Rj–Æb mK<Ö
3. zîR0zîmK<Ö
 - u(b7mAz mK<Ö
 - Qs•.ï_,mK<Ö
 - •èRÿËÿmK<Ö
4. `ËÿmK<Ö
 - •.}mK<Ö
 - S<R>mK<Ö
 - ^vSÑmK<Ö

8.2 mK<Öu(O<

RŸ€ýmK<Õu(O<ÿ

1. u(b7<¤<ÁmK<Õ

- kc^8v{ _U
- _ ^8v{ _U
- [Æx 'í•n
- O <Ý{it

2. [¢g {it mK<Õ

- R ^ú[¢g
- fôe°[¢g
- R –d[¢g
- gâ<â[¢g

3. <þz [fN`mK<Õ

- <þz R ^ú
- <þz [fN`
- •Û^i•ýŽ*
- b ~é<Ä[š

4. KPI€ h8mK<Õ

- e¹hHR ^ú
- € h8bg^L
- ~Ógœ<j{—
- b¥TJu b

8.3 mK<Õepcn

mK<ÕepcnQ/ÆY ÿ

1. u(b7mK<Õepcn

- {it TX•&b7
- N;{i•&b7
- [¢g •&b7
- •¹uL•&b7

2. O N mK<Õepcn

- kc^8O N
- <Õu(g O N
- •Çg O N

3. N RjmK<Õepcn

- <þz epcn
- <Ýg/epcn
- € h8epcn

8.4 mK<Õ]âQw

mK<Õ]âQw• béÿ

1. Jest - SUQCmK<ÕhFg¶
2. Testing Library - ~ÄNömk<Õ
3. Playwright - E2EmK<Õ
4. k6 - ``ÉÿmK<Õ

{,N]zà •è•r•Ð~ô

9.1 •è•rmAz

•è•rkešǎÿ

1. Năx g„^ú
 - O••V[%^Å
 - Năx • <Ñ
 - •Dn•O S
2. s¯Xf‘M•n
 - s¯XfSØ‘î<¾•n
 - ‘M•ne‡Nöu b
 - [Æ”¥lèQe
3. epcn^“•Áyû
 - epcn^“•þc¥
 - •Áyû• g,bg~L
 - epcnR YÉS
4. g RıT/R”
 - Pe^•hÄgå
 - g RıT/R”
 - ••}WG^a‘M•n
5. •è•ršŒ<Á
 - RŸËÿšŒ<Á
 - ``ÉÿšŒ<Á
 - [%QhšŒ<Á

9.2 s¯Xf‘M•n

‘M•n~ÿ<ôf ÿ

1. epcn^““M•n
 - DATABASE_URL: epcn^“•þc¥N2
 - SUPABASE_URL: Supabaseg RıW0W@
 - SUPABASE_ANON_KEY: S?T [Æ”¥

- SUPABASE_SERVICE_ROLE_KEY: g R_i[Æ"¥

2. <¤<Á'M•n

- JWT_SECRET: JWT[Æ"¥

- SESSION_SECRET: O <Ý[Æ"¥

3. g R_i'M•n

- PORT: g R_izĩSă

- NODE_ENV: •ĐˆLsˆXf

9.3 vÑc\$TJ<f

vÑc\$~y<¾<jÿ

1. g R_ir¶|` vÑc\$

- g R_i[Xm;hĂgă

- Peˆ.r¶|` hĂgă

- •Dn•O•u(vÑc\$

2. ``Éýc h vÑc\$

- TÍˆ"eö•ö vÑc\$

- T T 'İvÑc\$

- • <is‡vÑc\$

3. eă_xvÑc\$

- • <İeă_xe6—Æ

- <¿•İeă_xR g•

- [i<j eă_xOÝ[X

4. N R_jc h vÑc\$

- u(b7m;•Ăˆ|

- RÝÉyO•u(s‡

- _ ^8ˆLN:hĂmK

9.4 •Đ~ômAz

eă^8•Đ~ôÿ

1. eă_xR g•

- • <İeă_xc'gă

- ``ÉýtöˆˆR g•

- u(b7ˆLN:R g•

2. epcnY Ný

- [šg •êRˆY Ný

- Y NýŠŒ<Ă

- `bY o ~Ă

3. [%Qhfôe°
 - O••Vfôe°
 - [%Qh~eN
 - o m OïY

4. `€ýO S
 - `€ývÑcš
 - tö~[šOM
 - O S [že½

{,SAzà –D_U

10.1 b€g/• W<<ôf

RMzihFg¶• béNext.jsv„SŸVàÿ

1. g Rjzin2gÔe/c ÿ cÐSG™–\O`€ý
2. App Router•ïu1|û~ßÿ _ SÑOSš€Y}
3. Reactu ` b qŸÿ y>S:m;•Ã
4. TypeScript/c [€U„ÿ |{W<[%Qh

UI~ÄNö^“• béshadcn/uiv„SŸVàÿ

1. WúNŽRadix UIÿ Sı<¿•“Y}
2. Tailwind CSS–Æb ÿ h7_ pum;
3. ~ÄNö•(ıšØÿ <¾<¡•Ž%oÃ
4. Sı[šR6`'_:ÿ f NŽ~ôbα

epcn^“• béPostgreSQLv„SŸVàÿ

1. RŸ€ý_:Y'ÿ e/c JSON0 Qhe‡d }“{
2. `€ýz3[šÿ ~ı•Ç••g š€<Á
3. SupabaseX{ıÿ •Ð~ôb g,ON
4. RLSe/c ÿ [%Qhg:R6[€U„

10.2 g/<ı<ôf

g/<ı[šNiÿ

- KPI: Key Performance Indicatorÿ Qs•.~éeHc h
- RLS: Row Level Securityÿ ^L~š[%Qh{Vue
- JWT: JSON Web Tokenÿ JSON•Q~ÜNärL
- API: Application Programming Interfaceÿ ^"u(z ^c¥Sã
- ORM: Object Relational Mappingÿ [ù€EaQs|ûf \
- CDN: Content Delivery Networkÿ Q...[ıR SÑ•Q~Ü
- SSR: Server Side Renderingÿ g Rjzin2gÓ
- CSR: Client Side Renderingÿ [¢b7zın2gÓ

10.3 •)ue<î'h

•)ue<í	Qhyð	<ôf	
API	Application Programming Interface	^"u(z ^•c¥Sã	
JWT	JSON Web Token	JSON•Q~ÜNärL	
KPI	Key Performance Indicator	Qs•.~éeHc h	
ORM	Object Relational Mapping	[ùÆaQs	ûf \
RLS	Row Level Security	^L~\$[%Qh{Vue	
SSR	Server Side Rendering	g R;zin2gÓ	
CSR	Client Side Rendering	[¢b7zin2gÓ	
CDN	Content Delivery Network	Q...[¹R SÑ•Q~Ü	

10.4 rHg,S†Sò

rHg,<°_Uÿ
V1.0.0 - R YËrHg,(2024^t)
RŸ€yn SUÿ
- Wúx@hFg†|d-^ú
- u(b7<¤<Á|û~ß
- O N {it RŸ€ý
- [¢g {it RŸ€ý
- Wù<-N- ãRŸ€ý
- KPI€ h8RŸ€ý
- <Ÿg/^"RŸ€ý
- wâ<Æ^"RŸ€ý
- AIR©bKRŸ€ý
- epcnR g•RŸ€ý
-]ãSU{it RŸ€ý
- •"Ri{it RŸ€ý

10.5 g*ge%oÄR

T ~îrHg,<ijR ÿ
V1.1.0 - RŸ€ýXž_:rH
- yûR"zï• 'MO S
- epcn[üQúRŸ€ýXž_:
- b¥^hRŸ€ýbìU
- by'ÍdíO\O S
- • wâ|û~ßSG~§
V1.2.0 - fz€ýS SG~§rH
- AÍ€ýR>Xž_:
- •êR`S c`fP

- fz€ýc'sí
- ~„mKR g•
- fz€ý[¢g

V2.0.0 - ^sSðS rHg,
- Y O N e/c
- _ e>API^sSð
- cÖNð|û~ß
- •ê[šNI]åO\mA
- epcnN-Sð

e‡hc• R6ý x SÑVâ–
• R6eåg ý 2024^t
e‡hcrHg,ý V1.0
[ih8r¶` ý]ò[ih8